

- This exam contains 12 pages (including the front and back of this cover page) and 14 problems. Check to see if any pages are missing.
- This is the second exam and will count for 20% of your final grade. You have 120 minutes (5:00 pm - 7:00 pm) to complete this exam. You may use one index card of notes and a calculator on this exam, but no other outside sources may be consulted.

The following rules apply:

- Every problem on this exam, other than true-or-false problems, requires work of some form. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. **A correct answer that has no supporting work will not receive full credit.**
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 100 points total in this exam, as well as 2 bonus questions worth a total of 6 possible points.

Page:	3	4	5	6	7	8	9	10	Total
Points:	20	10	10	12	10	10	10	18	100
Score:									

Bonus:	
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1. Mark each statement as ☐ True or ☐ False by completely shading in the box corresponding to your answer.

2 pts

(a) If $f(x) = 3(5)^{x-2} + 2$, then $f(7) = 9377$.

☐ True☐ False

2 pts

(b) The logarithmic equation $\log_7(142) = a$ is equivalent to the exponential equation $7^a = 142$.

☐ True☐ False

2 pts

(c) The exponential equation $e^x = y$ is equivalent to the logarithmic equation $\ln(x) = y$.

☐ True☐ False

2 pts

(d) $\log_\pi \left(\sqrt{x^5 y^{-10}} \right) = 5/2 \log_\pi(x) - 5 \log_\pi(y)$.

☐ True☐ False

2 pts

(e) The angle $5\pi/4$ in radians is equivalent to the angle 225° in degrees

☐ True☐ False

2 pts

(f) The angle between 0 and 2π in radians that is coterminal to the angle $-3\pi/4$ is $5\pi/4$.

☐ True☐ False

2 pts

(g) $\sin(\pi/3) = \sqrt{2}/2$.

☐ True☐ False

2 pts

(h) $\sec\left(\frac{\pi}{3}\right) = 2$

☐ True☐ False

2 pts

(i) If $\sec(x) = 4.5$, then $\sec(-x) = -4.5$.

☐ True☐ False

2 pts

(j) $\csc(x)$ is the inverse function of $\sin(x)$.

☐ True☐ False

5 pts

2. Use the definition of a logarithm to solve the equation

$$12 - 2\log_{10}(8x + 4) = 8.$$

Ⓐ $x = 3$

Ⓑ $x = e^4$

Ⓒ $x = 12$

Ⓓ $x = 0$

5 pts

3. Solve the equation for x :

$$\ln(7) - \ln(7 - 7x) = \ln(8).$$

Ⓐ $x = 7/8$

Ⓑ $x = e^8$

Ⓒ $x = -7/8$

Ⓓ $x = 12$

5 pts

4. Find the length of the arc of a circle of radius 10 inches subtended by a central angle of 50° .

Ⓐ $25/9$ inches

Ⓑ π inches

Ⓒ $25/9\pi$ **inches**

Ⓓ $9/25\pi$ inches

5 pts

5. Find the area of a sector of a circle with radius 8 inches subtended by a central angle of $\pi/2$ radians.

Ⓐ 2π inches²

Ⓑ 32π inches²

Ⓒ 8π inches²

Ⓓ 16π **inches²**

7 pts

6. Find the (x, y) -coordinates of the point on a circle with radius 25 corresponding to an angle of $\frac{\pi}{6}$.

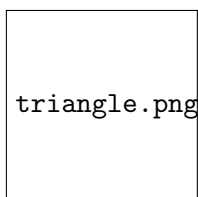
Ⓐ $\left(\frac{25}{2}, \frac{25\sqrt{3}}{2}\right)$ Ⓑ $\left(\frac{25\sqrt{3}}{2}, \frac{25}{2}\right)$ Ⓒ $(25\sqrt{3}, 25)$ Ⓓ $\left(\frac{25\sqrt{2}}{2}, \frac{25\sqrt{2}}{2}\right)$

5 pts

7. If $x \sin(x) = -4.5$, find $(-2x) \sin(-x)$.

10 pts

8. Solve the unknown sides of the triangle below:



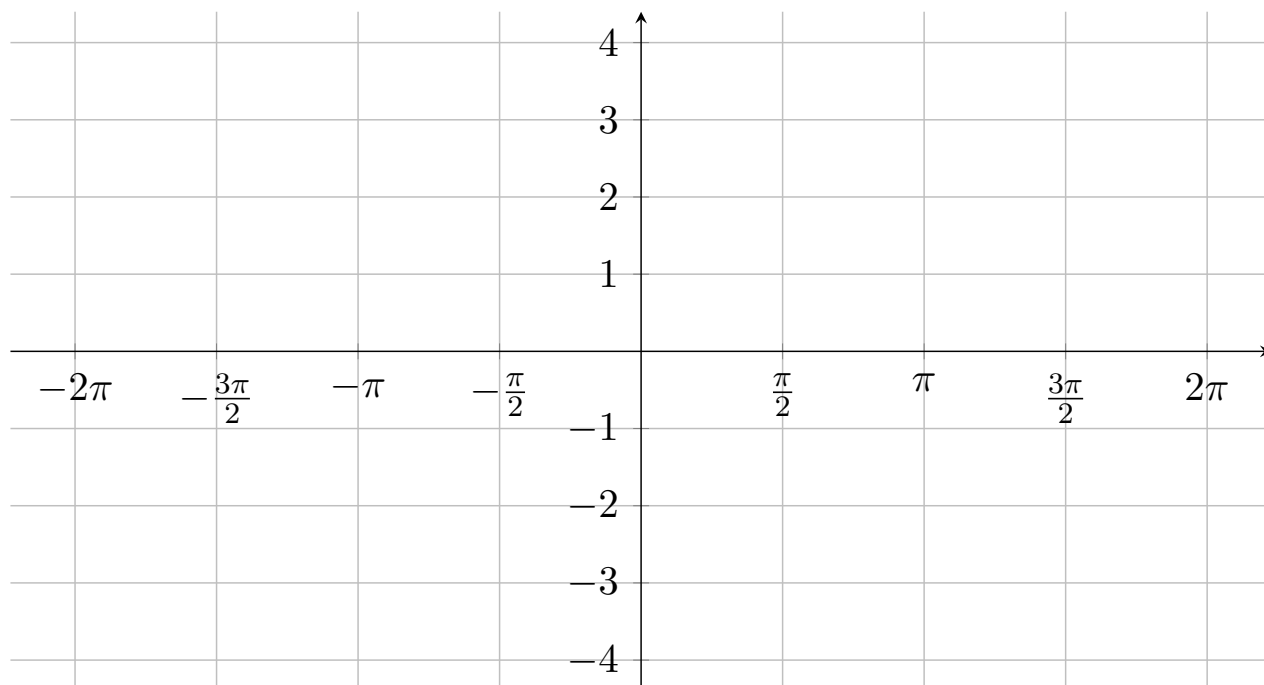
Ⓐ $a = \frac{11\sqrt{3}}{3}, c = \frac{22\sqrt{3}}{3}$

Ⓒ $a = \frac{11\sqrt{2}}{2}, c = \frac{11\sqrt{2}}{2}$

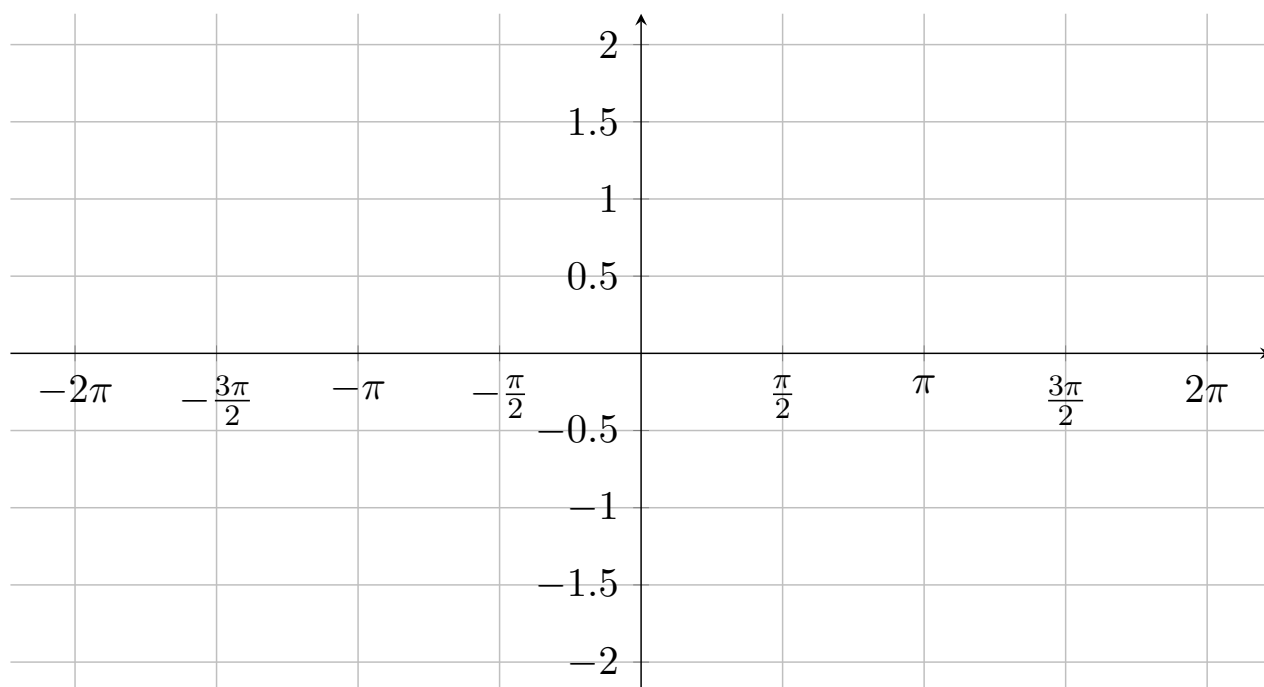
Ⓑ $a = 11, c = 22$

Ⓓ $a = 11, c = \frac{22\sqrt{3}}{3}$

10 pts

9. Graph the function $f(x) = 4 \cos(x)$.

10 pts

10. Graph the function $f(x) = 2 \sin(2x)$.

8 pts

11. State the amplitude, period, phase shift and vertical translation of the function

$$f(x) = 3 \sin \left(\frac{1}{3}x + \frac{3\pi}{4} \right) - 2$$

Amplitude : _____

Period : _____

Phase Shift : _____

Vertical Translation : _____

10 pts

12. Find all values of x in the interval $[0, 2\pi)$ such that $\cos(x) = -\frac{1}{2}$.

Ⓐ $x = 2\pi/3$ **and** $x = 4\pi/3$

Ⓒ $x = \pi/3$ and $x = 5\pi/3$

Ⓑ $x = \sqrt{2}/2$ and $x = -\sqrt{2}/2$

Ⓓ $x = 8\pi/3$ and $x = 10\pi/3$

3 (bonus)

13. The angle of elevation to the top of a building is found to be 15° from the ground at a distance of 1 mile from the base of the building. Find the height of the building to the nearest foot. (Hint: 1 mile is 5280 feet)

3 (bonus)

14. The London Eye is a huge Ferris wheel with a diameter of 135 meters (443 feet). It completes one rotation every 30 minutes. Riders board from a platform 2 meters above the ground. Express a rider's height above ground as a function of time in minutes.